Virtual Memory – MCQ Practice

10 Multiple-Choice Questions | Medium Difficulty

Topics: Virtual Memory, Page-Replacement Algorithms, Demand Paging, Dirty Bit, Address Translation and TLB

Instructions: Select the single best answer for each question. The answer key and explanations are provided at the end.

1. A system uses a 32-bit virtual address and a page size of 4 KB. How many bits are required for the page offset?

A. 10 bits
B. 12 bits
C. 20 bits
D. 22 bits

2. A process generates a page fault during demand paging. Which sequence best describes the normal handling of a valid page that is not currently in RAM?

A. Terminate the process immediately
B. Load the page from secondary storage into a free frame or replace a page, then update the page table
C. Clear the TLB and resume without loading the page
D. Write the entire process back to disk

3. A page-replacement algorithm uses the dirty bit. A victim page has dirty bit = 1. What must generally happen before its frame can be reused?

A. The page must be copied to the TLB
B. The page must be written back to disk if its changes are not already stored there
C. The page must be deleted from the executable file
D. The page must be marked as read-only

4. A process accesses pages in this order: 1, 2, 3, 1, 4. It has three empty frames and uses FIFO replacement. Which page is evicted when page 4 is accessed?

A. Page 1
B. Page 2
C. Page 3
D. No page is evicted

5. What is the primary purpose of a Translation Lookaside Buffer (TLB)?

A. To store modified pages permanently
B. To cache recently used virtual-to-physical address translations
C. To replace the page table completely in every system
D. To detect all page faults before memory access

6. A TLB has a hit ratio of 90%. A TLB lookup takes 10 ns, and a memory access takes 100 ns. Assuming one memory access on a TLB hit and two on a miss, what is the effective access time?

A. 110 ns
B. 120 ns
C. 130 ns
D. 210 ns

7. Which statement best explains why demand paging can improve memory utilization?

A. It loads every page before a process starts
B. It loads pages only when they are first needed
C. It eliminates the need for page tables
D. It guarantees that page faults never occur

8. A system uses LRU page replacement with three frames. The reference string is 1, 2, 3, 1, 4. Which page is replaced when page 4 is accessed?

A. Page 1
B. Page 2
C. Page 3
D. Page 4

9. A virtual address is divided into a page number and an offset. The page table maps virtual page 5 to physical frame 12. If the page size is 1 KB and the offset is 100 bytes, what is the physical address?

A. 5,220
B. 12,388
C. 12,100
D. 12,3880

10. A process repeatedly accesses a working set of pages larger than the available physical frames. Which outcome is most likely?

A. The TLB hit ratio must become 100%
B. The process may experience excessive page faults and thrashing
C. The dirty bit is automatically cleared on every access
D. Address translation becomes unnecessary

Answer Key and Explanations

1. Answer: B — 12 bits

4 KB = 4096 bytes = 2^12 bytes. Therefore, the page offset requires 12 bits.

2. Answer: B — Load the page from secondary storage into a free frame or replace a page, then update the page table

The OS locates the page, obtains a frame, loads the page, updates the page table, and restarts the faulting instruction.

3. Answer: B — The page must be written back to disk if its changes are not already stored there

A dirty page has been modified in memory. Its changes generally need to be written back to backing storage before the frame is reused.

4. Answer: A — Page 1

Pages 1, 2, and 3 fill the frames in that order. FIFO evicts the page that entered memory first: page 1.

5. Answer: B — To cache recently used virtual-to-physical address translations

The TLB is a fast cache that stores recent address translations, reducing the need to access the page table in memory.

6. Answer: B — 120 ns

Hit time = 10 + 100 = 110 ns. Miss time = 10 + 200 = 210 ns. EAT = 0.9 × 110 + 0.1 × 210 = 120 ns.

7. Answer: B — It loads pages only when they are first needed

Demand paging avoids loading pages that a process never accesses, reducing initial memory usage and unnecessary I/O.

8. Answer: B — Page 2

Before page 4 arrives, page 1 was most recently used, followed by page 3, then page 2. Page 2 is least recently used and is evicted.

9. Answer: B — 12,388

A 1 KB page contains 1024 bytes. Physical address = frame number × page size + offset = 12 × 1024 + 100 = 12,388.

10. Answer: B — The process may experience excessive page faults and thrashing

When the active working set cannot fit in available memory, pages are repeatedly evicted and reloaded, potentially causing thrashing.